

Youyin Hu

youyinhu6@gmail.com | <https://youyinhu.com>

Education

Master of Science in Operations Research and Control Theory

University of Shanghai for Science and Technology

Sep. 2024 – Present

Shanghai, China

- GPA: 4.0/4.5
- **Relevant Courses:** Numerical Methods for Differential Equations (98/100), Statistical Learning Theory (97/100), Graph Neural Networks (93/100), Stochastic Control (91/100)

Bachelor of Science in Applied Mathematics

University of Shanghai for Science and Technology

Sep. 2020 – Jun. 2024

Shanghai, China

- **Relevant Courses:** Ordinary Differential Equations (93/100), Probability Theory (95/100), Abstract Algebra (97/100), Machine Learning (90/100), Fuzzy Mathematics (95/100)

Research Experience

Gradient Tracking-Based Push-Sum Distributed Optimization Under Encoding-Decoding Scheme (*In Preparation*)

Independent Researcher

Apr. 2026 – Present

- Proposed a gradient tracking-based push-sum algorithm integrated with an encoding-decoding quantization scheme to achieve exact and communication-efficient distributed optimization over directed graphs.
- Designed an encoding-decoding mechanism that converts continuous gradient information into quantized data with provably bounded error, significantly reducing communication overhead in multi-agent networks.
- Incorporated gradient tracking techniques to correct the gradient bias arising from heterogeneous local objective functions, enabling exact convergence to the global optimum.
- Established rigorous convergence guarantees for the proposed algorithm over time-varying directed graphs, characterizing the impact of quantization error on convergence rate.

Push-Sum Distributed Optimization for Multi-Agent Systems Under Encoding-Decoding Scheme

Independent Researcher

May. 2025 – Apr. 2026

- Conceptualized and developed a novel homomorphic encryption-based push-sum framework to bridge the gap between data privacy and communication efficiency in multi-agent systems.
- Designed a unified protocol integrating Paillier encryption and encoding-decoding quantization, effectively handling asymmetric directed communication topologies.
- Derived rigorous theoretical convergence guarantees for uniformly strongly connected time-varying directed graphs, establishing explicit convergence rate bounds.
- Formulated a comprehensive error propagation analysis, proving that quantization errors decay exponentially to ensure optimal solution convergence.
- Authored the manuscript: “Push-Sum Distributed Optimization for Multi-Agent Systems Under Encoding-Decoding Scheme”, submitted to *International Journal of Robust and Nonlinear Control* (*Under Review*).

Privacy-Preserving Distributed Recursive Filtering for State-Saturated Systems with Quantization Effects

Independent Researcher

Sep. 2024 – Feb. 2025

- Developed a distributed recursive filter for state-saturated systems, utilizing output mask functions to ensure data privacy and scaled uniform quantization to optimize communication efficiency in sensor networks.
- Derived the upper bound of filtering error covariance via Riccati-like equations, proving its boundedness over a finite horizon and recursively solving for the optimal gain matrix to balance accuracy and stability.

- Verified the algorithm’s robustness against nonlinear saturation through a three-tank system simulation in MATLAB.
- Published findings in *International Journal of Network Dynamics and Intelligence* (2025, <https://doi.org/10.53941/ijndi.2025.100012>).

The 22nd China Graduate Mathematical Contest in Modeling

Team Leader

Sep. 2025

- Led modeling efforts in a 3-person team, developing turbulence intensity calculation models using Richardson number and TKE methods with multi-source data fusion.
- Implemented 3D spatial turbulence mapping algorithms and prediction models with 100m horizontal and 50m vertical resolution; performed optimal flight route planning using **Python** and **MATLAB** with comprehensive visualizations.
- Completed research paper and computational framework within 72 hours; awarded **National Third Prize** for innovative data fusion approach and practical solutions for low-altitude flight safety.

Research Interests

- Distributed Optimization
- Privacy and Secure Computing
- Network Communication Efficiency

Publications

1. Hu, Y.; Zhang, C.; Liu, S. **Privacy-Preserving Distributed Recursive Filtering for State-Saturated Systems with Quantization Effects.** *International Journal of Network Dynamics and Intelligence* 2025, 4(2), 100012. <https://doi.org/10.53941/ijndi.2025.100012>
2. Hu, Y.; Liu, S.; Wang, L. **Push-Sum Distributed Optimization for Multi-Agent Systems Under Encoding-Decoding Scheme.** Submitted to *International Journal of Robust and Nonlinear Control* (*Under Review*)
3. Hu, Y.; Liu, S. **Gradient Tracking-Based Push-Sum Distributed Optimization Under Encoding-Decoding Scheme.** (*In Preparation*)

Honors & Awards

National Third Prize , The 22nd China Graduate Mathematical Contest in Modeling <i>China</i>	2025
Outstanding Dissertation , University of Shanghai for Science and Technology <i>Shanghai, China</i>	2024

Teaching Experience

Teaching Assistant, Stochastic Control

Spring 2025

University of Shanghai for Science and Technology

- Assisted Prof. Shuai Liu with course instruction for graduate-level stochastic control.
- Delivered tutorial sessions on control theory fundamentals and applications.

Skills

Programming & Analysis: Python

Mathematical Software: MATLAB, Mathematica, Maple

Scientific Writing: LaTeX

Languages: Chinese (Native), English (IELTS 7.0)